Aman Sharma

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Education

- **The University of Edinburgh** *Sep 2022 - Jun 2026*
  BEng Computer Science(Hons) | Edinburgh, United Kingdom Grade classification: First Class
  – Recent Modules: Algorithmic Game Theory, Probabilistic Modeling, Advanced NLP, Discrete Maths and Probability, Foundational Data Science, Computer Systems, Functional Programming.
- **The University of Pennsylvania** *Jul 2024 - May 2025*
  Academic Exchange | Philadelphia, Pennsylvania, United States of America GPA: 3.85
  – Selected as 1 of 4 students for year-long Ivy League exchange. **Completed 9.5 credits** across the School of Engineering and Wharton, exceeding required course load.
  – Modules: Operating Systems, Computer Architecture, Compilers, Distributed Computing, Database Systems
  – Computer & Network Security, Machine Learning, Statistical Computing, Pipeline Optimisation for ML.

Experience

- **Bloomberg** *Jun 2025 - Sep 2025*
  Software Engineering Intern London
  – Redesigned the **FX auditing system** under the Electronic Trading team using Kafka.
  – Engineered for low-latency, distributed environments with strict performance and reliability constraints.
  – Built integration/regression tests to achieve and maintain >**90% code coverage** across critical components.
  – Ensured scalability to handle **2× current trading volume** while meeting a 50% latency improvement target.
- **Cadence Design Systems** *Jun 2024 - Sep 2024*
  Software Engineering Intern Edinburgh
  – Built new features for the flagship software Virtuoso for circuit design.
  – Utilised C++, Python (Flask), Node.JS, Linux, Bash Scripting, and Electronic Design Automation (EDA) tools.
  – **Built C++/Python pipelines** to expose runtime metrics to external systems, enabling automated monitoring.
- **EUFS (Edinburgh University Formula Student)** *Sep 2023 - Dec 2024*
  Simulation Software Developer Edinburgh
  – Developed simulation tech in the **7x(Best Simulation Award)** AI class winning Formula Student UK team.
  – Utilised C++, Node.JS, React.JS, Foxglove and the ROS set of tools to extend on existing technologies.

Personal Projects

- **Pipelined RISC-V Processor** *Nov 2024 - Dec 2024*
  [SystemVerilog, C] ***Github***
  – Built a 5-stage, **RV32IM pipelined RISC-V processor** with branch prediction, bypassing and stalling.
  – Integrated direct-mapped write-back cache (I$ / D$) to handle variable-latency memory accesses.
  – Achieved timing-closed synthesis and **FPGA deployment** for live demos showcasing system performance.
- **UNIX-like Operating System Simulator** *May 2025 - Jul 2025*
  [C, POSIX Threads, Custom Shell, FAT File System, Signals] ***Github***
  – Engineered an operating system, with **multithreaded process scheduling**, file system support, and a shell.
  – Created a light file system and a UNIX-like shell supporting job control, I/O redirection, and background tasks.
  – Built a scheduler using timer interrupts to simulate context switching and user-level **thread management**.
- **Ray Tracing Engine** *Jul 2024*
  [C++, SDL] ***Github***
  – Implemented a **physics-based renderer** from scratch using ray tracing equations to model light interactions.
  – Developed mathematical models for materials, objects and textures to support extensible scene design.
  – Accelerated rendering through GPU computation, SDL visualization, and **multithreaded architecture**.

Technical Skills and Interests

Languages: C++, C, Python, Java, Javascript, HTML/CSS, SQL, oCaml, R, x86 Assembly, Verilog, Solidity
Developer Tools: VS Code, JetBrains, GDB, Valgrind, Git, Bash, LaTeX, CI/CD, Linux, AWS, Docker
Technologies/Frameworks: NodeJS, MongoDB, ReactJS, TensorFlow, PyTorch, Kafka, Pandas, NumPy

Positions

- **Morgan Stanley, McKinsey & Co Spring Intern,** Received return offer *Apr 2024*
- **Junior Analyst,** Edinburgh University Trading and Investment Club *Sep 2023 - Jun 2024*

Research

- **Computing Clearing Payments in Financial Networks** *Sep 2025 - Present*
  Dissertation Edinburgh
  – Implementing and benchmarking algorithms to compute clearing payments under cascading financial defaults.
  – Analyzing trade-offs in convergence speed, numerical stability, and scalability across large simulated networks.